PRE-ALGEBRA (7TH) · UNIT 9: SPATIAL THINKING

Skate Park Designer

Use angles, polygons, circles, and transformations to design a town skate park.

CP2

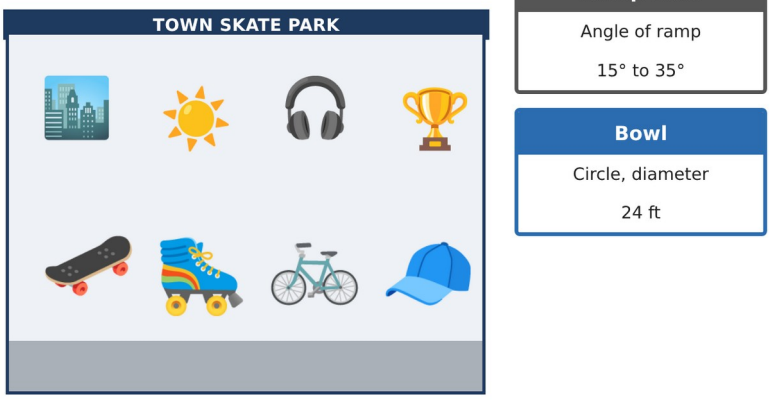
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The town council approved a new skate park and wants students to design it. The ramps need exact angles, the rails run parallel, the deck is a polygon, and the bowl is a circle.

You will use angle relationships, polygons, circles, and transformations to create a blueprint and a logo for the park.



Learning Goals

- I can identify complementary, supplementary, and vertical angles.
- I can find angle measures with parallel lines.
- I can use the circumference formula and transformations.

Project Timeline

When	What you will do
Day 1	Ramp Angles
Day 2	Rails & Deck
Day 3	Bowl & Logo
Final	Skate Park Blueprint + reflection

Part 1: Ramp Angles

Day 1 · A ramp meets the ground. The angle under the ramp and the angle on top of the ground line form a straight line.

Feature Cards

Quarter Pipe	Fun Box	Rail Pair
Ramp angle 30°	Regular hexagon top	Two parallel rails cut by a path

Bowl	Pyramid	Logo
Circle, d = 24 ft	Square base, 4 triangle sides	Must have rotational symmetry

1. The quarter pipe ramp angle is 30°. What is the supplementary angle on the other side?

   Starter: *30 + x = 180*

2. A support post makes a right angle with the ground. The ramp makes 30° with the ground. What angle does the ramp make with the post?

3. Two rails cross and make a 70° angle. What are the other three angles?

4. Name one pair of complementary and one pair of supplementary angles you found.

Part 2: Rails & Deck

Day 2 · The two rails are parallel. A path crosses them.

5. The path makes a 65° angle with the first rail. What is the corresponding angle at the second rail?

6. What is the same-side interior angle to the 65° angle?

7. The fun box top is a regular hexagon. Find the sum of its angles.

8. Find the measure of each angle of the regular hexagon.

Part 3: Bowl & Logo

Day 3 · Use π ≈ 3.14.

9. The bowl is a circle with diameter 24 ft. Find its circumference.

10. Translate the point (2, 3) on the logo 4 units right and 1 unit down.

11. Draw a logo with at least one line of symmetry. Show the line.

Final Product: Skate Park Blueprint

Your final product must include:

- Blueprint with the ramp, rails, fun box, and bowl
- All angle measures labeled
- Hexagon angle sum
- Bowl circumference
- Your logo with symmetry

Reflection

R1. How do parallel rails help you find angles without measuring?

R2. Why does a regular hexagon have 120° angles?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Angles, polygons, circle measures, and transformations are correct and produce a buildable design. Goes beyond the minimum with a creative, realistic choice.	Angles, polygons, circle measures, and transformations are correct and produce a buildable design.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Blueprint is labeled with angles and measurements and includes the logo and calculations. Polished, organized, and visually engaging.	Blueprint is labeled with angles and measurements and includes the logo and calculations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.